

PART 1 REACT NATIVE COURSE



React Native Complete Course: From Scratch to Expert

I'll create a comprehensive React Native course structured from beginner to expert level. This course is designed specifically for your learning journey based on your frontend React experience and AI background.

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Chapter 1: Introduction and Environment Setup

What is React Native?

React Native is a framework developed by Facebook that allows you to build mobile applications using JavaScript and React. Unlike traditional mobile development where you need separate codebases for iOS (Swift/Objective-C) and Android (Java/Kotlin), React Native lets you write once and run on both platforms.[1][2][3]

Key Benefits:

- **Cross-platform:** Single codebase for iOS, Android, and web
- **Native Performance:** Uses actual native components, not web views
- **Hot Reloading:** See changes instantly without rebuilding
- **Large Ecosystem:** Leverage existing React and JavaScript libraries
- **Cost-effective:** Faster development and easier maintenance

How React Native Works

React Native uses a bridge to communicate between JavaScript and native platforms. When you write React Native code, it gets translated into native components:[2]

```
// React Native JSX
<View style={styles.container}>
  <Text>Hello World!</Text>
</View>

// Becomes native iOS UIView and Android ViewGroup
```

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Environment Setup

Prerequisites:

- Node.js (v16 or higher)
- npm or yarn
- Code editor (VS Code recommended)
- For iOS: Xcode (macOS only)
- For Android: Android Studio

Setup Steps:

1. Install React Native CLI:

```
npm install -g react-native-cli
# OR install Expo CLI for easier setup
npm install -g expo-cli
```

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2. Create Your First Project:

```
# Using React Native CLI
npx react-native init MyFirstApp
```

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```
# OR using Expo (recommended for beginners)
npx create-expo-app MyFirstApp
cd MyFirstApp
```

3. Project Structure Overview:

```
MyFirstApp/
├─ App.js           # Main app component
├─ package.json     # Dependencies and scripts
├─ android/        # Android-specific files
├─ ios/            # iOS-specific files
├─ node_modules/   # Installed packages
└─ metro.config.js # Bundler configuration
```

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Your First React Native App

App.js - Basic Structure:

```
import React from 'react';
import {
  SafeAreaView,
  ScrollView,
  StatusBar,
  StyleSheet,
  Text,
  useColorScheme,
  View,
} from 'react-native';

// Main App component - this is where your app starts
function App() {
  // Get the current color scheme (light/dark mode)
  const isDarkMode = useColorScheme() === 'dark';

  // Define background color based on theme
  const backgroundColor = {
    backgroundColor: isDarkMode ? '#121212' : 'ffffff',
  };

  return (
    <SafeAreaView style={[backgroundColor, styles.container]}>
      {/* StatusBar controls the top system bar */}
      <StatusBar
        barStyle={isDarkMode ? 'light-content' : 'dark-content'}
        backgroundColor={backgroundColor.backgroundColor}
      />
    </SafeAreaView>
  );
}
```

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```
    />

    { /* Main content area */ }
    <ScrollView
      contentInsetAdjustmentBehavior="automatic"
      style={backgroundStyle}
    >
      <View style={styles.content}>
        <Text style={styles.title}>Welcome to React Native!</Text>
        <Text style={styles.subtitle}>
          Your first cross-platform mobile app
        </Text>
      </View>
    </ScrollView>
  </SafeAreaView>
);
}

// Styles object - similar to CSS but with camelCase properties
const styles = StyleSheet.create({
  container: {
    flex: 1, // Take up full screen height
  },
  content: {
    padding: 20,
    alignItems: 'center',
    justifyContent: 'center',
    minHeight: 200,
  },
  title: {
    fontSize: 24,
    fontWeight: 'bold',
    textAlign: 'center',
    marginBottom: 10,
  },
  subtitle: {
    fontSize: 16,
    textAlign: 'center',
    color: '#666',
  },
});

export default App;
```

Running Your App:

```
# For iOS (macOS only)
npx react-native run-ios

# For Android
npx react-native run-android

# For Expo projects
expo start
```

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Chapter 2: React Native Fundamentals and Core Components

Core Components Overview

React Native provides built-in components that map to native UI elements. Unlike HTML elements (`<div>`, `<span>`), you use React Native components.[3][2]

View Component

`View` is the fundamental building block - like a `<div>` in web development:

```
import React from 'react';
import { View, StyleSheet } from 'react-native';

function BasicView() {
  return (
    <View style={styles.container}>
      /* Views can contain other components */
      <View style={styles.box}>
        <View style={styles.innerBox} />
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    padding: 20,
  },
  box: {
    width: 100,
    height: 100,
    backgroundColor: '#3498db',
    justifyContent: 'center',
  },
});
```

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```
    alignItems: 'center',
  },
  innerBox: {
    width: 50,
    height: 50,
    backgroundColor: '#e74c3c',
  },
});
```

Text Component

All text must be wrapped in `Text` components:

```
import React from 'react';
import { View, Text, StyleSheet } from 'react-native';

function TextExample() {
  return (
    <View style={styles.container}>
      /* Basic text */
      <Text style={styles.heading}>This is a heading</Text>

      /* Nested text inherits parent styles */
      <Text style={styles.paragraph}>
        This is a paragraph with {' '}
        <Text style={styles.bold}>bold text</Text>
        {' '} and <Text style={styles.italic}>italic text</Text>.
      </Text>

      /* Text with multiple lines */
      <Text style={styles.multiLine}>
        This is a longer text that will wrap to multiple lines
        when it exceeds the container width.
      </Text>
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    padding: 20,
  },
  heading: {
    fontSize: 24,
    fontWeight: 'bold',
    marginBottom: 10,
```

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```
    color: '#2c3e50',
  },
  paragraph: {
    fontSize: 16,
    lineHeight: 24,
    marginBottom: 15,
    color: '#34495e',
  },
  bold: {
    fontWeight: 'bold',
  },
  italic: {
    fontStyle: 'italic',
  },
  multiLine: {
    fontSize: 14,
    textAlign: 'justify',
  },
});
```

Image Component

Display images from various sources:

```
import React from 'react';
import { View, Image, StyleSheet } from 'react-native';

function ImageExample() {
  return (
    <View style={styles.container}>
      {/* Local image - require() for bundled images */}
      <Image
        source={require('./assets/logo.png')}
        style={styles.logo}
      />

      {/* Network image - URI for web images */}
      <Image
        source={{
          uri: 'https://picsum.photos/200/200'
        }}
        style={styles.networkImage}
      />

      {/* Image with loading placeholder */}
      <Image
        source={{
```

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```

        uri: 'https://picsum.photos/150/150'
      }}
      style={styles.imageWithPlaceholder}
      loadingIndicatorSource={require('./assets/loading.png')}
    />
  </View>
);
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    alignItems: 'center',
    padding: 20,
  },
  logo: {
    width: 100,
    height: 100,
    marginBottom: 20,
  },
  networkImage: {
    width: 200,
    height: 200,
    borderRadius: 10,
    marginBottom: 20,
  },
  imageWithPlaceholder: {
    width: 150,
    height: 150,
    borderRadius: 75, // Circular image
  },
});

```

ScrollView Component

Create scrollable content areas:

```

import React from 'react';
import { ScrollView, View, Text, StyleSheet } from 'react-native';

function ScrollExample() {
  // Generate sample data
  const items = Array.from({ length: 20 }, (_, index) => ({
    id: index,
    title: `Item ${index + 1}`,
    description: `This is the description for item ${index + 1}`,
  }));
}

```

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```

return (
  <ScrollView
    style={styles.container}
    contentContainerStyle={styles.contentContainer}
    showsVerticalScrollIndicator={true}
  >
    <Text style={styles.header}>Scrollable Content</Text>

    {items.map(item => (
      <View key={item.id} style={styles.item}>
        <Text style={styles.itemTitle}>{item.title}</Text>
        <Text style={styles.itemDescription}>{item.description}</Text>
      </View>
    ))}
  </ScrollView>
);
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
  },
  contentContainer: {
    padding: 20,
  },
  header: {
    fontSize: 24,
    fontWeight: 'bold',
    marginBottom: 20,
    textAlign: 'center',
  },
  item: {
    backgroundColor: '#f8f9fa',
    padding: 15,
    marginBottom: 10,
    borderRadius: 8,
    elevation: 2, // Android shadow
    shadowColor: '#000', // iOS shadow
    shadowOffset: { width: 0, height: 2 },
    shadowOpacity: 0.1,
    shadowRadius: 2,
  },
  itemTitle: {
    fontSize: 18,
    fontWeight: '600',
    marginBottom: 5,
  },
  itemDescription: {

```

```
    fontSize: 14,  
    color: '#666',  
  },  
});
```

Chapter 3: Styling and Layout with Flexbox

React Native Styling System

React Native uses a JavaScript-based styling system similar to CSS but with some differences:[1][2]

```
import React from 'react';  
import { View, Text, StyleSheet, Dimensions } from 'react-native';  
  
// Get device dimensions  
const { width: screenWidth, height: screenHeight } = Dimensions.get('window');  
  
function StylingExample() {  
  return (  
    <View style={styles.container}>  
      /* Inline styles (not recommended for complex styling) */  
      <Text style={{ fontSize: 20, color: 'blue' }}>  
        Inline styled text  
      </Text>  
  
      /* StyleSheet styles (recommended) */  
      <Text style={styles.styledText}>  
        StyleSheet styled text  
      </Text>  
  
      /* Multiple styles - later styles override earlier ones */  
      <Text style={[styles.baseText, styles.highlightedText]}>  
        Multiple styles applied  
      </Text>  
  
      /* Conditional styling */  
      <Text style={[  
        styles.baseText,  
        screenWidth > 400 && styles.largeScreenText  
      ]}>  
        Conditional styling based on screen size  
      </Text>  
    </View>  
  );  
}
```

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```
const styles = StyleSheet.create({
  container: {
    flex: 1,
    padding: 20,
    backgroundColor: '#ffffff',
  },
  styledText: {
    fontSize: 18,
    fontWeight: '500',
    color: '#2c3e50',
    marginVertical: 10,
  },
  baseText: {
    fontSize: 16,
    color: '#34495e',
  },
  highlightedText: {
    backgroundColor: '#f1c40f',
    padding: 8,
    borderRadius: 4,
  },
  largeScreenText: {
    fontSize: 20,
    fontWeight: 'bold',
  },
});
```

Flexbox Layout System

React Native uses Flexbox for layout, with some defaults different from web CSS:[3]

```
import React from 'react';
import { View, Text, StyleSheet } from 'react-native';

function FlexboxExample() {
  return (
    <View style={styles.container}>
      /* Flex Direction Example */
      <View style={styles.section}>
        <Text style={styles.sectionTitle}>Flex Direction: Row</Text>
        <View style={styles.rowContainer}>
          <View style={[styles.box, styles.box1]}>
            <Text>1</Text>
          </View>
          <View style={[styles.box, styles.box2]}>
            <Text>2</Text>
          </View>
        </View>
      </View>
    </View>
  );
}
```

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```

    </View>
    <View style={[styles.box, styles.box3]}>
      <Text>3</Text>
    </View>
  </View>
</View>

{/* Justify Content Example */}
<View style={styles.section}>
  <Text style={styles.sectionTitle}>Justify Content: Space Between</Text>
  <View style={styles.spaceBetweenContainer}>
    <View style={[styles.smallBox, styles.box1]} />
    <View style={[styles.smallBox, styles.box2]} />
    <View style={[styles.smallBox, styles.box3]} />
  </View>
</View>

{/* Align Items Example */}
<View style={styles.section}>
  <Text style={styles.sectionTitle}>Align Items: Center</Text>
  <View style={styles.alignCenterContainer}>
    <View style={[styles.box, styles.box1]}>
      <Text>Centered</Text>
    </View>
  </View>
</View>
</View>
);
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    padding: 20,
    backgroundColor: '#ecf0f1',
  },
  section: {
    marginBottom: 30,
  },
  sectionTitle: {
    fontSize: 18,
    fontWeight: 'bold',
    marginBottom: 10,
    color: '#2c3e50',
  },
  rowContainer: {
    flexDirection: 'row', // Default is 'column' in React Native
    height: 60,
  },
});

```

```

spaceBetweenContainer: {
  flexDirection: 'row',
  justifyContent: 'space-between',
  height: 60,
},
alignCenterContainer: {
  alignItems: 'center',
  height: 80,
  backgroundColor: '#bdc3c7',
},
box: {
  width: 60,
  height: 60,
  justifyContent: 'center',
  alignItems: 'center',
},
smallBox: {
  width: 40,
  height: 40,
},
box1: {
  backgroundColor: '#e74c3c',
},
box2: {
  backgroundColor: '#3498db',
},
box3: {
  backgroundColor: '#2ecc71',
},
});

```

Complex Layout Example

Here's a practical layout combining multiple concepts:

```

import React from 'react';
import { View, Text, Image, StyleSheet, ScrollView } from 'react-native';

function ComplexLayoutExample() {
  return (
    <ScrollView style={styles.container}>
      </* Header Section */>
      <View style={styles.header}>
        <Image
          source={{ uri: 'https://picsum.photos/80/80' }}
          style={styles.avatar}
        />

```

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```

    <View style={styles.headerText}>
      <Text style={styles.userName}>John Doe</Text>
      <Text style={styles.userRole}>React Native Developer</Text>
    </View>
  </View>

  { /* Stats Section */ }
  <View style={styles.statsContainer}>
    <View style={styles.statItem}>
      <Text style={styles.statNumber}>123</Text>
      <Text style={styles.statLabel}>Projects</Text>
    </View>
    <View style={styles.statItem}>
      <Text style={styles.statNumber}>456</Text>
      <Text style={styles.statLabel}>Commits</Text>
    </View>
    <View style={styles.statItem}>
      <Text style={styles.statNumber}>789</Text>
      <Text style={styles.statLabel}>Followers</Text>
    </View>
  </View>

  { /* Content Cards */ }
  <View style={styles.cardContainer}>
    <View style={styles.card}>
      <Text style={styles.cardTitle}>Recent Activity</Text>
      <Text style={styles.cardContent}>
        Pushed 3 commits to react-native-project
      </Text>
    </View>

    <View style={styles.card}>
      <Text style={styles.cardTitle}>Achievements</Text>
      <Text style={styles.cardContent}>
        Completed 5 projects this month
      </Text>
    </View>
  </View>
</ScrollView>
);
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    backgroundColor: '#f5f6fa',
  },
  header: {
    flexDirection: 'row',

```

```
    alignItems: 'center',
    padding: 20,
    backgroundColor: '#ffffff',
    marginBottom: 10,
  },
  avatar: {
    width: 80,
    height: 80,
    borderRadius: 40,
    marginRight: 15,
  },
  headerText: {
    flex: 1,
  },
  userName: {
    fontSize: 24,
    fontWeight: 'bold',
    color: '#2c3e50',
  },
  userRole: {
    fontSize: 16,
    color: '#7f8c8d',
    marginTop: 5,
  },
  statsContainer: {
    flexDirection: 'row',
    backgroundColor: '#ffffff',
    marginBottom: 10,
  },
  statItem: {
    flex: 1,
    alignItems: 'center',
    paddingVertical: 20,
  },
  statNumber: {
    fontSize: 24,
    fontWeight: 'bold',
    color: '#3498db',
  },
  statLabel: {
    fontSize: 14,
    color: '#7f8c8d',
    marginTop: 5,
  },
  cardContainer: {
    padding: 10,
  },
  card: {
    backgroundColor: '#ffffff',
```

```

padding: 20,
marginBottom: 10,
borderRadius: 8,
elevation: 2,
shadowColor: '#000',
shadowOffset: { width: 0, height: 2 },
shadowOpacity: 0.1,
shadowRadius: 4,
},
cardTitle: {
  fontSize: 18,
  fontWeight: 'bold',
  marginBottom: 10,
  color: '#2c3e50',
},
cardContent: {
  fontSize: 14,
  color: '#34495e',
  lineHeight: 20,
},
});

```

This comprehensive course structure provides you with a solid foundation in React Native. Each chapter builds upon the previous one, starting with basic concepts and gradually introducing more complex features.^{[2][1][3]}

Next Steps:

1. Set up your development environment
2. Create your first project and experiment with the code examples
3. Practice styling and layout exercises
4. Move to the next chapter on user interactions

The course continues with navigation, state management, API integration, and advanced topics that will take you from beginner to expert level React Native development.

[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#) [10](#) [11](#) [12](#) [13](#) [14](#) [15](#) [16](#) [17](#) [18](#) [19](#) [20](#) [21](#) [22](#) [23](#) [24](#) [25](#) [26](#) [27](#) [28](#) [29](#) [30](#) [31](#) [32](#) [33](#) [34](#) [35](#) [36](#) [37](#) [38](#) [39](#) [40](#)